

Session-13: Collection Processing & Transformation Operations –

13a: Java Priority Queue –

Program:

```
import java.util.*;
import java.io.*;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }

    public double getCGPA() {
        return cgpa;
    }
}

class Priorities {
    public List<Student> getStudents(List<String> events) {
        PriorityQueue<Student> pq = new PriorityQueue<>(
            Comparator.comparing(Student::getCGPA).reversed()
                .thenComparing(Student::getName)
                .thenComparing(Student::getID)
        );
```

```

for (String event : events) {
    String[] parts = event.split(" ");
    if (parts[0].equals("ENTER")) {
        String name = parts[1];
        double cgpa = Double.parseDouble(parts[2]);
        int id = Integer.parseInt(parts[3]);
        pq.add(new Student(id, name, cgpa));
    } else if (parts[0].equals("SERVED")) {
        pq.poll();
    }
}

List<Student> remaining = new ArrayList<>();
while (!pq.isEmpty()) {
    remaining.add(pq.poll());
}
return remaining;
}
}

public class Solution {
    private final static Scanner scan = new Scanner(System.in);
    private final static Priorities priorities = new Priorities();
    public static void main(String[] args) {
        int totalEvents = Integer.parseInt(scan.nextLine());
        List<String> events = new ArrayList<>();
        while (totalEvents-- != 0) {
            String event = scan.nextLine();
            events.add(event);
        }
        List<Student> students = priorities.getStudents(events);
        if (students.isEmpty()) {
            System.out.println("EMPTY");
        }
    }
}

```

```

    } else {
        for (Student st : students) {
            System.out.println(st.getName());
        }
    }
}
}

```

Output:

The screenshot displays the output of a Java program across three panes:

- Left Pane (Test Results):** Shows a list of test cases from 0 to 6. All test cases are marked as passed with a green checkmark and a lock icon.
- Middle Pane (Compiler Message):** Displays the message "Success". Below it, the "Input (stdin)" for test case 0 is shown:


```

1 12
2 ENTER John 3.75 50
3 ENTER Mark 3.8 24
4 ENTER Shafaet 3.7 35
5 SERVED
6 SERVED
7 ENTER Samiha 3.85 36
8 SERVED
9 ENTER Ashley 3.9 42

```
- Right Pane (Expected Output):** Shows the "Expected Output" for test case 0:


```

8 SERVED
9 ENTER Ashley 3.9 42
10 ENTER Maria 3.6 46
11 ENTER Anik 3.95 49
12 ENTER Dan 3.95 50
13 SERVED

```

13b: Java ArrayList –

Program:

```

import java.io.*;
import java.util.*;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        String s = sc.nextLine();

        ArrayList<ArrayList<String>> a = new ArrayList<ArrayList<String>>(n);
    }
}

```

```

for (int i = 0; i < n; i++) {
    s = sc.nextLine();
    a.add(i, new ArrayList<String>(Arrays.asList(s.split("\\s"))));
}

int m = sc.nextInt();
for (int i = 0; i < m; i++) {
    int x = sc.nextInt();
    int y = sc.nextInt();
    if (x <= a.size() && y < a.get(x-1).size() && y >= 0) {
        System.out.println(a.get(x-1).get(y));
    } else {
        System.out.println("ERROR!");
    }
}
}
}

```

Output:

✓ Test case 0	1	5	✓ Test case 0	8	1 3
✓ Test case 1	2	5 41 77 74 22 44	✓ Test case 1	9	3 4
✓ Test case 2	3	1 12	✓ Test case 2	10	3 1
✓ Test case 3	4	4 37 34 36 52	✓ Test case 3	11	4 3
✓ Test case 4	5	0	✓ Test case 4	12	5 5
✓ Test case 5	6	3 20 22 33	Expected Output		
	7	5			
	8	1 3			
	9	3 4			
	10	3 1			
	11	4 3	✓ Test case 5	1	74
	12	5 5		2	52
				3	37
				4	ERROR!
				5	ERROR!